

ChE 311 – Transport Phenomena I –Fall 2025

Rubric Assessment Summary

The assessment of ABET Student Outcome 1 was conducted using the final exam as the primary direct assessment instrument. The exam was scored out of 100 points.

ABET Outcome 1

An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.

Performance Indicator (PI)	Level 3: Satisfactory >65	Level 2: Developing 45 to 65	Level 1: Unsatisfactory <45
1. Apply mathematical and scientific principles to formulate models and systems relevant to the subject	Able to successfully combines mathematical and/or scientific principles to formulate models and systems relevant to chemical engineering	Chooses a mathematical model or scientific principle that applies to an engineering problem, but has trouble in model development	Does not understand the connection between mathematical models and the system or process to be analyzed or designed
2. Solve engineering problems by using the concepts of algebra, calculus, differential equations and/or linear algebra	applies concepts of algebra, calculus, differential equations and/or linear algebra to solve chemical engineering problems	Shows nearly complete understanding of applications of calculus and/or linear algebra in problem-solving	Does not understand the application of algebra, calculus, differential equations and/or linear algebra in solving chemical engineering problems
6. The uses of appropriate resources needed to solve problems	Uses appropriate resources to locate information needed to solve problems	Uses limited resources to solve problems	Uses no resources to solve problems

FINAL EXAM TOTAL POINTS 100

Average Score	70.97
Standard dev.	17.19

RESULTS BY PERFORMANCE INDICATOR

Mapped SO	1
Average	2.56
Std Dev	0.56
S	21
D	14
U	1

Summary of Findings

The **average score** on the final exam was **70.97**, with a **standard deviation of 17.19**.

This indicates a **satisfactory level of overall student achievement**, with a relatively wide spread of scores, suggesting variability in student mastery of the assessed material.

The **average performance indicator score** for SO1 was **2.56 (on a 3-point scale)** with a **standard deviation of 0.56**.

This average exceeds the commonly used attainment benchmark of **2.0**, indicating that **SO1 is satisfactorily attained**.

- **Satisfactory (S): 21 students**
- **Developing (D): 5 students**
- **Unsatisfactory (U): 4 students**

Interpretation

The majority of students (21 out of 36) performed at or above the developing level, with more than half demonstrating strong proficiency.

Overall, the results suggest that **students are generally achieving Student Outcome 1**, particularly in their ability to apply knowledge and solve engineering-related problems. However, the relatively high standard deviation in exam scores highlights areas for improvement.

Recommended Improvements

To enhance student attainment:

- Additional instructional support may be needed for lower-performing students.
- Targeted review sessions or formative assessments could help reinforce key problem-solving skills.
- Further analysis of exam questions may identify specific concepts where students underperformed.

Conclusion

Based on the assessment data, **Student Outcome 1 is considered attained**, though **continuous improvement actions are recommended** to reduce performance variability and support students who are not meeting expectations.

EXAM 3

Closed Book (Refer to the provided equations sheet)

Time: 90 minutes

1. A long vertical cylinder with radius R is rotating with constant angular velocity Ω within a large body of fluid.

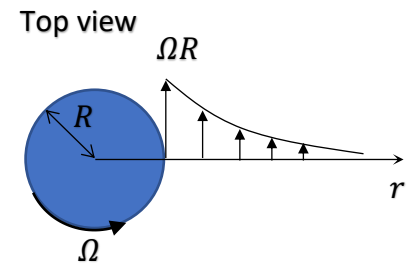
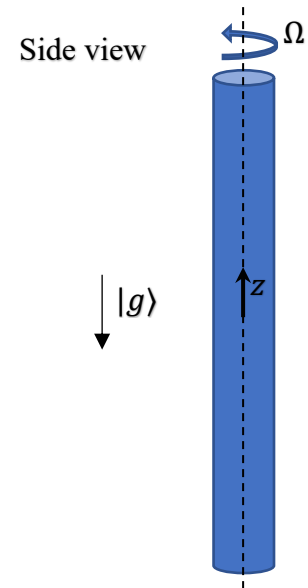
Postulates:

- Steady state, laminar flow.
- Newtonian fluid with constant viscosity μ and constant density ρ .
- Body of fluid around the cylinder is infinitely large.
- $v_r = 0$ and $v_z = 0$
- Pressure within the fluid is not a function of θ .

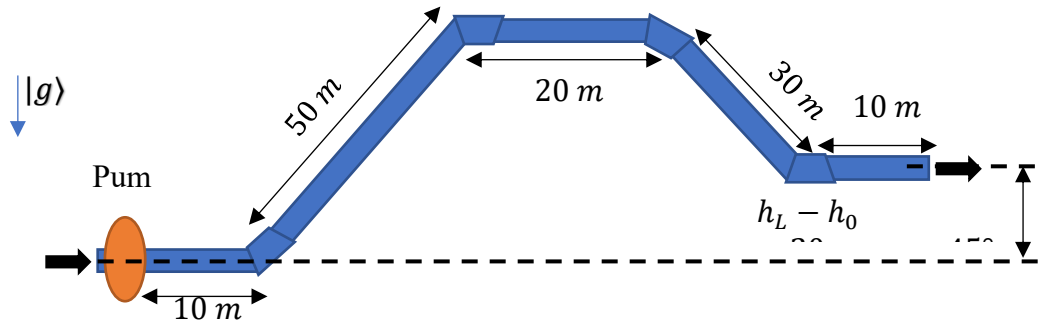
Answer the following questions in terms of parameters:

ρ , μ , Ω , and R .

- 1.1. Using the mass balance (equation of continuity), demonstrate that v_θ is not a function of θ .
- 1.2. Using the equation of motion, derive an ordinary differential equation (ODE) for v_θ . Integrate the ODE to get an equation for v_θ as a function of r . Write down two boundary conditions for v_θ and evaluate the integration constants.



2. Water is pumped between two reservoirs at the rate of $10 \frac{kg}{s}$. The pipeline consists of a pump, five smooth circular pipes of 20 cm inner radius (lengths of which are provided in the schematic below), and four 45° elbows. The resistance of each elbow is roughly equivalent to that offered by a pipe whose length is 15 diameters. The density of water is $\rho = 1000 \frac{kg}{m^3}$ and its viscosity is $\mu = 1.06 \text{ mPa}\cdot\text{s}$.



- 2.1. Using the provided chart, determine the Fanning friction factor for the above system.
- 2.2. If both reservoirs are at atmospheric pressure ($1.01 \times 10^5 \text{ Pa}$), what is the pressure at the pump outlet?

$$Re = \frac{\rho \langle v \rangle D}{\mu}$$

$$F_k \equiv (2\pi RL) \left(\frac{1}{2} \rho \langle v \rangle^2 \right) f = (\mathcal{P}_0 - \mathcal{P}_L) \pi R^2$$

1.1. (10 POINTS)

Continuity equation:

$$\frac{\partial \rho}{\partial t} - \operatorname{div} \rho \langle v \rangle = 0$$

For cylindrical coordinates, we use equation B.4-2. For constant density fluid with $v_r = 0$ and $v_z = 0$

$$\frac{1}{r} \frac{\partial \rho v_\theta}{\partial \theta} = 0 \rightarrow \frac{\partial v_\theta}{\partial \theta} = 0$$

1.2. (40 POINTS)

After eliminating the terms that are zero equation B.6-5 simplifies as:

$$0 = \frac{d}{dr} \left(\frac{1}{r} \frac{d(rv_\theta)}{dr} \right)$$

Above differential equation can be integrated twice, which results in:

$$v_\theta = C_1 r + \frac{C_2}{r}$$

B.C. $v_\theta(r = R) = \Omega R$

B.C. $v_\theta(r = \infty) = 0$

$$\begin{aligned} v_\theta(r = \infty) = 0 &\rightarrow C_1 = 0 \\ v_\theta(r = R) = \Omega R &\rightarrow C_2 = \Omega R^2 \\ v_\theta &= \Omega R^2 \frac{1}{r} \end{aligned}$$

2.1. (20 POINTS)

$$\langle v \rangle = \frac{w}{\rho \pi R^2} = \frac{10 \frac{Kg}{s}}{1 \times 10^3 \frac{kg}{m^3} \times \pi \times (0.2 \text{ m})^2} = 0.0796 \frac{m}{s}$$
$$Re = \frac{D \rho \langle v \rangle}{\mu} = \frac{0.4 \text{ m} \times 1 \times 10^3 \frac{kg}{m^3} \times 0.0796 \frac{m}{s}}{1.06 \times 10^{-3} \text{ Pa.s}} = 3.003 \times 10^4$$
$$f \left(Re = 3.003 \times 10^4, \frac{k}{D} = 0 \right) = 0.006$$

2.2. (30 POINTS)

$$L = (120 + 4 \times 15 \times 0.4) \text{ m} = 144 \text{ m}$$
$$\mathcal{P}_0 - \mathcal{P}_L = \frac{2fL\rho\langle v \rangle^2}{D} = \frac{2 \times 0.006 \times 144 \text{ m} \times 1 \times 10^3 \frac{kg}{m^3} \times \left(0.0796 \frac{m}{s}\right)^2}{0.4 \text{ m}} = 27.37 \text{ Pa}$$
$$\mathcal{P}_0 - \mathcal{P}_L = p_0 - p_L + \rho g(h_0 - h_L) = 27.37 \text{ Pa}$$
$$p_L = 1.01 \times 10^5 \text{ Pa}$$
$$p_0 = 27.37 \text{ Pa} + 1.01 \times 10^5 \text{ Pa} + 1.3859 \times 10^5 \text{ Pa} = 2.3962 \times 10^5 \text{ Pa}$$